YAZAN ELTOPGEY

Mechatronics & Automation Engineer | Robotics Specialist | AI Developer

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Mechatronics and Automation Engineer with expertise in autonomous robotics, industrial automation systems, AI, and IoT/IIoT solutions. Proficient in ROS2, PLC programming, reinforcement learning, and computer vision. Published researcher with proven success in award-winning autonomous and industrial systems.

EXPERIENCE

Mechatronics Engineer — Oct 2025 – Present

RCB, Jordan

- Programmed and maintained CNC machines and PLC RATIC systems for industrial automation workflows
- Built a full-stack project management website with database tracking tasks per employee and project
- Developed n8n automation pipeline for AI-powered invoice analysis from sales data

Supervisor Engineer — Al Fahd Quadruped Robot — 2025 – 2026

Al Hussein Technical University (HTU)

- Ported full ROS2 stack (URDF, launch files, configs) and integrated ros2_control for real-time joint control
- Achieved stable locomotion on real hardware — resolved all real-robot implementation gaps
- 1st Place — Best Graduation Project at HTU

Autonomous Service Robot Research — 2024 – June 2025

Graduation Project

- Developed hybrid navigation system combining reinforcement learning with traditional algorithms
- Implemented full sensor fusion (camera, LiDAR, encoders) with SLAM mapping on Raspberry Pi
- National Winner — Best Graduation Project in Robotics and AI (2025)

Robotics Research Intern — Jul 2024 – Oct 2024

India

- Published research paper: 'Autonomous Mobile Robot Navigation using RL with Nav2'
- Optimized ROS2 navigation stack reducing path planning computation time by 25%
- Developed reinforcement learning algorithms for dynamic environment navigation

Industrial IoT & Automation Intern — Aug 2023 – Jan 2024

INJO4.0, Jordan

- Implemented predictive maintenance systems reducing equipment downtime by 40%
- Automated data collection workflows processing 5000+ daily sensor readings
- Developed real-time monitoring dashboards using PLCs and Norvi IIoT devices

EDUCATION

Bachelor of Science in Mechatronics Engineering — September 2025

The Hashemite University, Zarqa, Jordan

SKILLS

Programming:	C++, Python, CUDA, MATLAB
Robotics:	ROS, ROS2, Gazebo, Isaac Sim, SLAM
AI/ML:	Machine Learning, Deep Learning, Reinforcement Learning, OpenCV, TensorFlow, PyTorch, YOLO
Hardware:	Arduino, Raspberry Pi, Jetson Nano, ESP32, STM32
Industrial:	PLC Programming, IoT, IIoT, SCADA, Industry 4.0
Design:	SolidWorks, AutoCAD, Fusion 360, Blender
Tools:	GitHub, Linux, Docker, n8n

PROJECTS

Al Fahd — 12-DOF Quadruped Robot | ROS2, ros2_control, C++
Real-world quadruped with full ROS2 integration and stable hardware locomotion. 1st Place HTU.

Autonomous Vehicle — CARLA | CARLA, ROS2, Computer Vision
Urban autonomous driving with traffic recognition and obstacle avoidance in simulation.

AI Smart Radar System | YOLOv8, DeepSORT, OCR
Dual-camera speed violation detection system with real-time vehicle tracking.

Industrial IoT & Automation | IIoT, PLC, SCADA
Real-time monitoring dashboards reducing equipment downtime by 40%.

Autonomous Service Robot | ROS2, RL, SLAM, CV
National award-winning autonomous navigation robot using RL and full sensor fusion.

Mobile Robot Navigation with RL | RL, TurtleBot3, LiDAR
Published research on autonomous navigation using reinforcement learning with Nav2.

Cancer Detection — Deep Learning | TensorFlow, PyTorch
High-accuracy cancer detection from medical images with deployable inference API.

Sumo Robot Competition | Arduino, IR Sensors
Autonomous sumo robot with opponent detection strategy for regional competition.